

Lecture 18 Worksheet

Custom NPU Dialect: ODS + Verifier + Test Design

Exercise A - Field Classification

Field	Operand / Type / Attribute / Verifier / Cost model	Reason
lhs tile		
rhs tile		
tile m/n/k		
lhs layout		
DMA elements		
DMA src/dst memory space		
barrier scope		
estimated cycles		

Exercise B - Fill ODS skeleton

```
def NPU_DmaOp : NPU_Op<"dma", [ /* traits/interfaces */ ]> {  
  let arguments = (ins /* source */, /* dest */, /* direction */, /* size */);  
  let results = (outs /* token */);  
  let assemblyFormat = "...";  
  let hasVerifier = 1;  
}
```

Exercise C - Negative diagnostics

Op	Illegal case	Expected diagnostic
npu.matmul	tile = 8x16x32	
npu.matmul	rhs_layout = mk	
npu.dma	elements = 0	
npu.barrier	empty token list	

Exercise D - FileCheck template

```
// RUN: npu-opt %s | FileCheck %s  
// RUN: npu-opt %s -verify-diagnostics  
// CHECK-LABEL: func.func @...  
// CHECK: npu.matmul  
// expected-error @+1 {{...}}
```